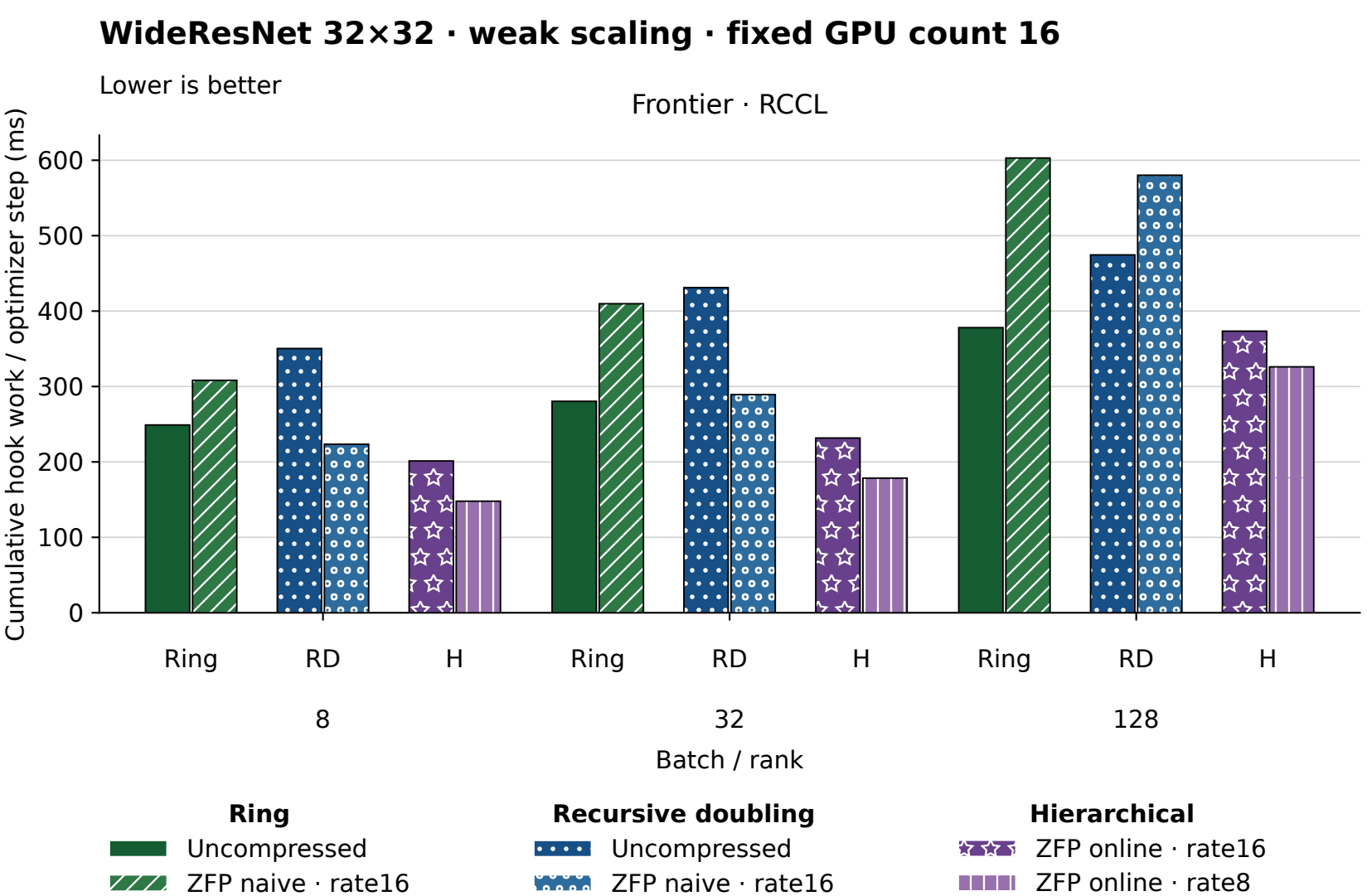

WideResNet 32×32 · weak scaling · fixed GPU count 16
Lower is better
Frontier · RCCL
Cumulative hook work / optimizer step (ms)
0
100
200
300
400
500
600
Ring
RD
H
Ring
RD
H
Ring
RD
H
8
32
128
Batch / rank
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate